

Experiment No. 3

Experiment Name

Implementation and Analysis of Spatial Filtering Techniques using Low-Pass and High-Pass Filters

Aim

To implement and analyze spatial domain filtering techniques using low-pass and high-pass filters for image smoothing, noise reduction, edge enhancement, and feature preservation using Python and OpenCV.

Purpose

Spatial filtering is one of the fundamental operations in digital image processing, used to enhance image quality by modifying pixel values based on their neighboring pixels. Low-pass filters are primarily used to reduce noise and smooth images, whereas high-pass filters emphasize edges, fine details, and abrupt intensity changes, which are essential for feature extraction and object recognition.

In this experiment, students will implement various spatial filtering techniques on real-world images and evaluate their impact on image quality, edge preservation, and noise removal. The experiment demonstrates the significance of filtering as a preprocessing step in computer vision applications such as medical imaging, surveillance, remote sensing, and autonomous systems.

Prerequisites

- Basic knowledge of digital image processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Understanding of image kernels and convolution operations
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required libraries and load a grayscale or color image for processing.
2. Display the original image and analyze its visual characteristics.
3. Apply Gaussian Blur to reduce image noise while preserving important structures.
4. Apply Median Filtering to remove impulse (salt-and-pepper) noise from the image.

5. Apply Average (Mean) Filtering and compare its smoothing performance with Gaussian filtering.
6. Implement Laplacian Filtering to enhance image edges and fine details.
7. Apply Sobel Edge Detection in both horizontal and vertical directions to detect image gradients.
8. Compare the outputs of different low-pass and high-pass filters in terms of noise reduction, edge preservation, and computational performance.
9. Display the original and filtered images along with suitable observations highlighting the strengths and limitations of each filtering technique.
10. Interpret the results and discuss the applications of spatial filtering in real-world computer vision systems.

Questions

1. What is spatial filtering? How is it used in digital image processing?
2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.
3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications.
4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?
5. Explain the role of convolution kernels in spatial filtering.
6. What is the purpose of the Sobel and Laplacian operators in edge detection?
7. Why are filtering operations considered an essential preprocessing step in computer vision?
8. Discuss the trade-off between image smoothing and edge preservation during filtering.
9. Mention four real-world applications where spatial filtering techniques are widely used.
10. Compare spatial domain filtering with frequency domain filtering in terms of implementation and practical applications.